

- 1) Find $\sin 2x$, $\cos 2x$, and $\tan 2x$ from given info.
 $\csc(x) = 8$, $\tan(x) < 0$

$$\frac{0\pi}{y} = \frac{8}{1}$$

$$x = \sqrt{8^2 - 1^2}$$

$$= \sqrt{63}$$

$$= -3\sqrt{7}$$

$$\sin(2x) = 2 \sin x \cos x$$

$$= 2 \left(\frac{1}{8} \right) \left(-\frac{3\sqrt{7}}{8} \right)$$

$$= \boxed{-\frac{3\sqrt{7}}{32}}$$

$$\cos(2x) = 1 - 2 \sin^2 x$$

$$= 1 - 2 \left(\frac{1}{8} \right)^2$$

$$= 1 - \frac{1}{32}$$

$$= \boxed{\frac{31}{32}}$$

$$\tan(2x) = \frac{\sin 2x}{\cos 2x}$$

$$= \frac{-\frac{3\sqrt{7}}{32}}{\frac{31}{32}}$$

$$= \boxed{-\frac{3\sqrt{7}}{31}}$$

- 2) Find $\sin 2x$, $\cos 2x$, and $\tan 2x$ from given info.

$$\sin(x) = -\frac{12}{13}, x \text{ in Quadrant III}$$



$$\sin(2x) = 2 \sin x \cos x$$

$$= 2 \left(-\frac{12}{13} \right) \left(-\frac{5}{13} \right)$$

$$= \boxed{\frac{120}{169}}$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= \left(-\frac{5}{13} \right)^2 - \left(-\frac{12}{13} \right)^2$$

$$= \frac{25}{169} - \frac{144}{169}$$

$$= \boxed{-\frac{119}{169}}$$

$$\tan(2x) = \frac{\sin 2x}{\cos 2x}$$

$$= \frac{\frac{120}{169}}{-\frac{119}{169}}$$

$$= \boxed{-\frac{120}{119}}$$

- 3) Use the formulas for lowering powers to rewrite the expression in terms of the first power of cosine

$$7 \cos^2 x \sin^4 x$$

$$= \boxed{7 \left(\frac{1 + \cos 2x}{2} \right) \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 - \cos 2x}{2} \right)}$$

- 4) Use half-angle formula to find exact value of expression
 $\cos(22.5^\circ)$

$$\cos\left(\frac{45}{2}\right) = \sqrt{\frac{1 + \cos(45^\circ)}{2}}$$

$$= \sqrt{\frac{1 + \frac{\sqrt{2}}{2}}{2}}$$

- 5) Use half-angle formula to find exact value of expression

$$\cos\left(\frac{9\pi}{8}\right)$$

$$\cos\left(\frac{9\pi}{8}\right) = -\sqrt{\frac{1 + \cos\left(\frac{9\pi}{4}\right)}{2}}$$

$$= -\sqrt{\frac{1 + \frac{\sqrt{2}}{2}}{2}}$$

- 6) Simplify the expression using a Double-Angle or Half-Angle Formula

a) $\cos^2(38^\circ) - \sin^2(38^\circ)$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\cos(2 \cdot 38^\circ) = \cos^2(38^\circ) - \sin^2(38^\circ)$$

$$= \boxed{\cos(76^\circ)}$$

b) $\cos^2(7\theta) - \sin^2(7\theta)$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\cos(2 \cdot 7\theta) = \cos^2(7\theta) - \sin^2(7\theta)$$

$$= \boxed{\cos(14\theta)}$$

7) Simplify the expression using a Double-Angle or Half-Angle Formula

a) $\frac{\sin(6^\circ)}{1 + \cos(6^\circ)}$

$$\tan\left(\frac{u}{2}\right) = \frac{\sin u}{1 + \cos u}$$

$$\tan\left(\frac{6^\circ}{2}\right) = \frac{\sin(6^\circ)}{1 + \cos(6^\circ)}$$

$$= \boxed{\tan(3^\circ)}$$

b) $\frac{1 - \cos(100^\circ)}{\sin(100^\circ)}$

$$\tan\left(\frac{u}{2}\right) = \frac{1 - \cos u}{\sin u}$$

$$= \tan\left(\frac{100^\circ}{2}\right)$$

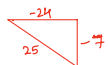
$$= \tan(50^\circ)$$

8) Find $\sin\left(\frac{x}{2}\right)$, $\cos\left(\frac{x}{2}\right)$, and $\tan\left(\frac{x}{2}\right)$ from given info

$$\cos x = -\frac{24}{25}, 180^\circ < x < 270^\circ$$

x is in quadrant III, $\frac{x}{2}$ is in quadrant II

$$90^\circ < \frac{x}{2} < 135^\circ$$



$$\sin\left(\frac{x}{2}\right) = \sqrt{\frac{1 - \cos x}{2}} = \sqrt{\frac{1 - \left(-\frac{24}{25}\right)}{2}}$$

$$\cos\left(\frac{x}{2}\right) = \sqrt{\frac{1 + \cos x}{2}} = -\sqrt{\frac{1 + \left(-\frac{24}{25}\right)}{2}}$$

$$\tan\left(\frac{x}{2}\right) = \frac{\sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right)} = \frac{\sqrt{\frac{1 - \left(-\frac{24}{25}\right)}{2}}}{-\sqrt{\frac{1 + \left(-\frac{24}{25}\right)}{2}}}$$

9) Find $\sin\left(\frac{x}{2}\right)$, $\cos\left(\frac{x}{2}\right)$, and $\tan\left(\frac{x}{2}\right)$ from given info

$$\tan(x) = \sqrt{2}, 0^\circ < x < 90^\circ$$

x is in quadrant I, $\frac{x}{2}$ is in quadrant I

$$0^\circ < \frac{x}{2} < 45^\circ$$



$$\sin\left(\frac{x}{2}\right) = \sqrt{\frac{1 - \cos x}{2}} = \sqrt{\frac{1 - \left(\frac{1}{\sqrt{3}}\right)}{2}}$$

$$\cos\left(\frac{x}{2}\right) = \sqrt{\frac{1 + \cos x}{2}} = \sqrt{\frac{1 + \left(\frac{1}{\sqrt{3}}\right)}{2}}$$

$$\tan\left(\frac{x}{2}\right) = \frac{\sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right)} = \frac{\sqrt{\frac{1 - \left(\frac{1}{\sqrt{3}}\right)}{2}}}{\sqrt{\frac{1 + \left(\frac{1}{\sqrt{3}}\right)}{2}}}$$

10) Write given expression as an algebraic expression in x .

$$\tan(2 \sin^{-1}(x))$$

$$\theta = \sin^{-1}x$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$



$$\tan 2\theta = \frac{2 \left(\frac{x}{\sqrt{1-x^2}} \right)}{1 + \left(\frac{x}{\sqrt{1-x^2}} \right)^2}$$

11) Find the exact value of the given expression.

$$\sin\left(2 \cos^{-1}\left(\frac{8}{17}\right)\right)$$



$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \left(\frac{15}{17} \right) \left(\frac{8}{17} \right)$$

$$= \boxed{\frac{240}{289}}$$

12) Evaluate the expression under the given conditions.

$$\cos(2\theta); \sin(\theta) = -\frac{24}{25}, \theta \text{ in QIII}$$



$$\cos(2\theta) = 1 - 2 \sin^2 \theta$$

$$= 1 - 2 \left(-\frac{24}{25} \right)^2$$

$$= \boxed{-\frac{527}{625}}$$

13) Write the product as a sum
 $\sin(x) \sin(7x)$

$$= \frac{1}{2} [\cos(x - 7x) - \cos(x + 7x)]$$

$$= \frac{1}{2} [\cos(-6x) - \cos(8x)]$$

14) Write the product as a sum
 $7 \cos(8x) \cos(11x)$

$$= (7) \left(\frac{1}{2} \right) [\cos(8x + 11x) + \cos(8x - 11x)]$$

$$= \frac{7}{2} [\cos(19x) + \cos(-3x)]$$

15) Write the sum as a product
 $\sin(x) - \sin(4x)$

$$= 2 \cos\left(\frac{x+4x}{2}\right) \sin\left(\frac{x-4x}{2}\right)$$

$$= 2 \cos\left(\frac{5x}{2}\right) \sin\left(-\frac{3x}{2}\right)$$

16) Write the sum as a product
 $\cos(15x) + \cos(8x)$

$$= 2 \cos\left(\frac{15x+8x}{2}\right) \cos\left(\frac{15x-8x}{2}\right)$$

$$= 2 \cos\left(\frac{23x}{2}\right) \cos\left(\frac{7x}{2}\right)$$

17) Find the exact value of the product.

$$7 \cos(37.5^\circ) \cos(7.5^\circ)$$

$$= (7) \left(\frac{1}{2} \right) [\cos(37.5^\circ + 7.5^\circ) + \cos(37.5^\circ - 7.5^\circ)]$$

$$= \frac{7}{2} [\cos(45^\circ) + \cos(30^\circ)]$$

$$= \frac{7}{2} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \right)$$

$$= \frac{7}{2} \left(\frac{\sqrt{2} + \sqrt{3}}{2} \right)$$

$$= \frac{7(\sqrt{2} + \sqrt{3})}{4}$$

18) Prove the identity.

$$\sin(10x) = 2 \sin(5x) \cos(5x)$$

Start with the right-hand side of the identity. Apply a Double-Angle Formula and simplify.

$$2 \sin(5x) \cos(5x) = \sin\left(2 \cdot \left(\boxed{5x} \quad \checkmark\right)\right)$$

$$= \sin\left(\boxed{10x} \quad \checkmark\right)$$

19) Prove the identity

$$4(\sin^6(x) + \cos^6(x)) = 4 - 3\sin^2(2x)$$

$$\text{LHS} = 4[(\sin^2 x)^3 + (\cos^2 x)^3]$$

$$= 4[(\sin^2 x + \cos^2 x)(\sin^4 x - \sin^2 x \cos^2 x + \cos^4 x)]$$

$$= 4[(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x - \sin^2 x \cos^2 x]$$

$$= 4(1 - 3\sin^2 x \cos^2 x)$$

$$= 4 - 3(4\sin^2 x \cos^2 x)$$

$$= 4 - 3\sin^2(2x)$$

$$= \text{RHS}$$

20) Use a Sum-to-Product Formula to show the following.

$$\cos(100^\circ) - \cos(200^\circ) = \sin(50^\circ)$$

$$\cos(100^\circ) - \cos(200^\circ) = -2 \sin\left(\frac{100^\circ + 200^\circ}{2}\right) \sin\left(\frac{100^\circ - 200^\circ}{2}\right) \quad \text{Sum-to-Product Formula}$$

$$= -2 \sin(150^\circ) \sin\left(\frac{-100^\circ}{2}\right) \quad \text{Simplify fractions}$$

$$= -2\left(\frac{1}{2}\right) \sin(-50^\circ) \quad \text{Evaluate sine}$$

$$= -\sin(-50^\circ) \quad \text{Multiply}$$

$$= \sin(50^\circ) \quad \text{Even-Odd identity and simplify}$$